Federated Small Language Models for Personalized and Interpretable Women's Health Support

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ABSTRACT

The digitalization of women's health is expanding the collection of hormonal and physiological measurements and self-reported observations through mobile applications, wearable devices, and connected health platforms. However, translating these heterogeneous data into personalized, understandable insights while protecting sensitive records remains a challenge. This paper proposes a federated small language model (SLM) framework for menstrual phase prediction and natural-language feedback generation without centralizing raw health records. We experiment with Qwen2.5-3B-Instruct and Phi-3-mini-Instruct, two SLMs whose quantized versions have been shown to run on smartphones in prior work. Using the mcPHASES dataset, we convert hormonal, physiological, and self-reported features into structured feature-to-text representations for fine-tuning with low-rank adaptation (LoRA). Each participant is treated as a client, with locally trained LoRA adapters combined through federated averaging while raw records remain local. The framework predicts the menstrual, follicular, fertility, and luteal phases and generates concise feedback based on the supplied features. The federated Qwen2.5-3B and Phi-3-mini models achieve accuracies of 0.839 and 0.825 and macro F1-scores of 0.818 and 0.803, respectively. Qualitative examples illustrate the generated feedback. On a workstation, phase prediction takes approximately 1–2 s and feedback generation 5–8.5 s. These findings support the potential of federated SLMs for decentralized interpretation of menstrual health measurements and personalized feedback in digital health applications.

1. Introduction

The digitalization of women's health is expanding the collection and interpretation of physiological, behavioral, and self-reported measurements through wearable sensors, smartwatches, smart rings, mobile applications, and connected health platforms. These technologies support symptom tracking, physiological monitoring, reproductive health management, and personalized wellness interventions in everyday settings [1–3]. Artificial intelligence further extends these measurement workflows by transforming collected data into predictions and individualized health insights [4]. As illustrated in Fig. 1, measurements and observations collected through wearable devices and smartphone-based platforms can provide complementary evidence of changes relevant to menstrual health. However, translating these heterogeneous data into personalized, understandable feedback while protecting sensitive health information remains a challenge. This motivates digital measurement workflows that connect multimodal health data with predictive interpretation and user-facing explanations, while reducing reliance on centralized processing [4, 5].

Prior user studies have shown that women use menstrual tracking applications not only for prediction but also to understand their bodies. However, users often struggle to interpret their logged observations and the resulting predictions, expressing a need for transparent, personalized, and context-aware feedback grounded in their symptoms and physiological signals [6, 7]. Recent multimodal studies similarly highlight users' interest in interpreting multiple physiological signals together, alongside the limited support that existing systems provide for this process [8]. These findings reveal a gap between collecting health measurements and translating them into understandable, individualized insights.

In this work, we investigate menstrual phase prediction and feedback generation as a representative application of digitalization in women's health measurement workflows. Menstrual health involves interconnected physiological, hormonal, and behavioral changes that can affect physical symptoms, emotional well-being, and daily functioning [9]. Hormonal and physiological measurements, complemented by self-reported observations, therefore provide a meaningful basis for computational interpretation. Although mobile applications and wearable devices increasingly support menstrual monitoring, existing systems primarily emphasize symptom logging, cycle estimation, and period prediction [10]. Their outputs often provide limited explanation of how individual symptoms, hormonal indicators, and behavioral observations relate to the predicted phase [11]. Advancing these workflows requires methods that connect heterogeneous health data with both phase predictions and understandable, feature-grounded feedback.

Protecting these data is an additional requirement for the digitalization of menstrual health monitoring. Applications may collect sensitive symptom reports, hormonal measurements, physiological signals, and daily behavioral information [12]. Cloud-based processing can require transmitting these data to centralized servers [11], raising concerns about privacy, trust, data governance, and potential misuse of reproductive health information [13]. Federated learning offers a means of training models across distributed datasets while retaining raw records locally. However, its integration with language-model-based interpretation and personalized feedback remains an important research direction for menstrual health support. Cloud-hosted language models also introduce dependence on data transmission and remote

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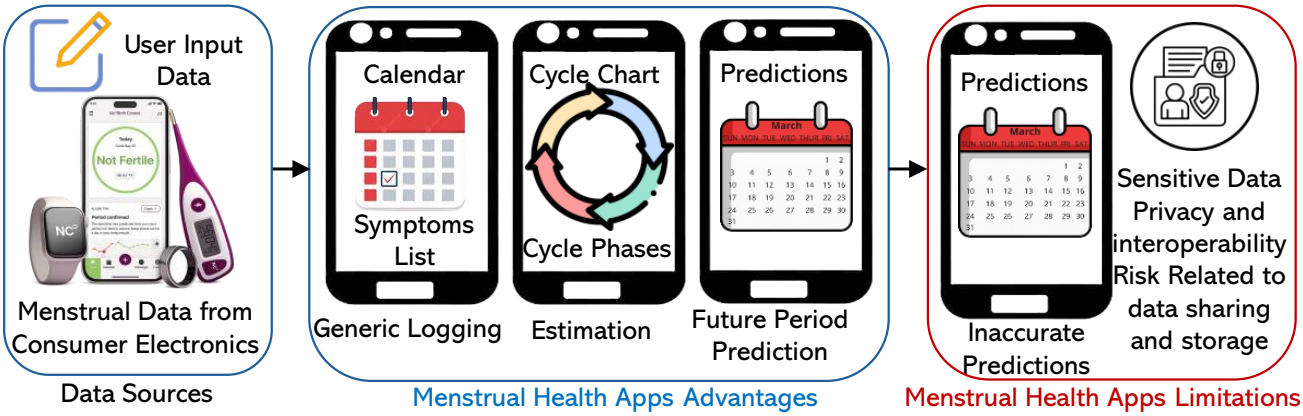


Figure 1: Existing menstrual health applications on mobile devices do not jointly address personalized feedback generation and privacy preservation.

computational resources [14]. These challenges motivate a framework that combines decentralized model adaptation with the interpretation of menstrual health measurements and observations to generate personalized feedback [15].

In this paper, we investigate how federated small language models (SLMs) can transform heterogeneous menstrual health measurements and self-reported observations into phase predictions and personalized natural-language feedback, while retaining raw records locally during training. We propose a federated SLM framework that connects distributed health data with predictive interpretation and user-facing feedback, supporting the digitalization of menstrual health monitoring workflows. Combining SLMs with federated learning offers a potential approach to collaborative model adaptation without centralizing sensitive user records [16].

The compact size of SLMs also motivates their consideration for edge and mobile inference, although deployment requirements depend on model precision, runtime optimization, and device capabilities. Beyond prediction, SLMs can express relationships between input features and model outputs in accessible natural language [17]. This capability may help users interpret their recorded symptoms and physiological measurements. Our study examines this approach through federated fine-tuning, predictive evaluation, qualitative feedback analysis, and workstation inference measurements. Our main contributions are as follows:

- We propose a federated SLM framework for interpreting menstrual health measurements and self-reported observations, treating each participant as a client and retaining raw records locally during training.
- We design a feature-to-text representation that converts hormonal, physiological, and self-reported features into structured prompts for menstrual phase prediction and personalized natural-language feedback generation.
- We develop a federated LoRA-based fine-tuning approach for Qwen2.5-3B-Instruct and Phi-3-mini-Instruct, enabling collaborative adaptation through the exchange of lightweight adapter parameters rather than full model weights or raw records.
- We evaluate predictive performance, user-level behavior, and workstation inference resource usage, qualitatively examine generated feedback, and discuss a pathway to smartphone deployment informed by prior on-device demonstrations.

The remainder of this paper is organized as follows. Section 2 reviews related work, Section 3 presents the proposed framework, Section 4 reports the results, Section 5 outlines a pathway to smartphone deployment, Section 6 discusses limitations and future work, and Section 7 concludes the paper.

2. Related Work

Our work lies at the intersection of consumer electronics-based women's health tracking, federated learning for privacy preservation, and language-model-driven personalized feedback generation. In this section, we review existing approaches in these areas and explain how our work differs by integrating privacy-preserving learning with personalized feedback generation for consumer electronics environments.

2.1. Sensor-Based Consumer Electronics for Women's Health Tracking

Commercial mobile apps [18–24] and wearable devices [25–28] collect physiological, behavioral, and self-reported data, enabling more personalized health monitoring over time. Among these applications, menstrual tracking has received particular attention for monitoring cycle patterns, symptoms, hormonal changes, and overall well-being. Although many existing apps support period prediction, symptom logging, and fertility estimation, they offer limited interpretability and do not fully use sensor data to provide deeper, more personalized insights.

Most health-tracking apps focus on tracking and prediction rather than on explaining their outputs to users. They

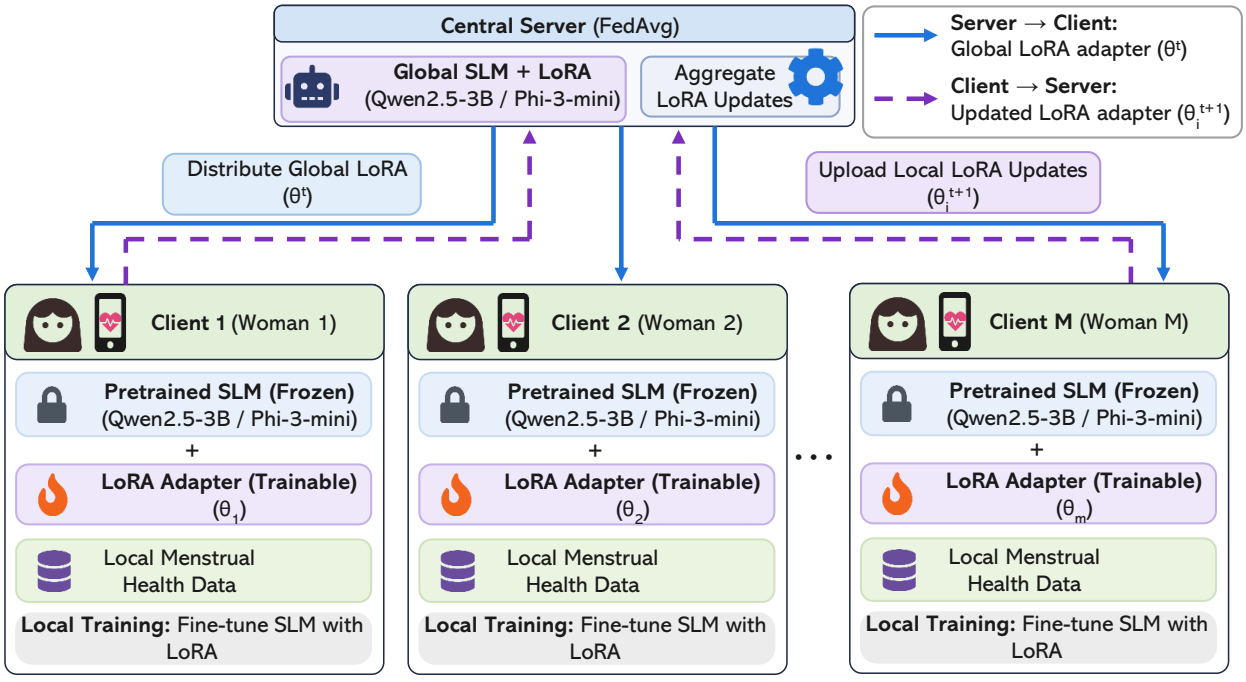


Figure 2: Federated LoRA-based fine-tuning of small language models for menstrual health intelligence.

also typically do not offer faithful, feature-grounded natural-language feedback that clearly links reported symptoms and physiological signals to the output. In addition, many of these systems rely heavily on self-reported data, which can be incomplete, inconsistent, or noisy. Privacy is another major concern, as menstrual health data are highly sensitive and require careful handling. Furthermore, advanced features are often restricted to subscription plans or tied to ecosystem-specific wearable devices, which limits accessibility and broader adoption.

2.2. Language Model-Based Feedback Generation for Women's Health Data

Language-based feedback generation transforms women's health data into clear, personalized natural-language feedback that supports health understanding and daily decision-making. Recker et al. [29] discussed how large language models (LLMs), particularly ChatGPT, can support women's health care in gynecology and obstetrics by improving access to understandable health information. They conclude that such tools can help users learn about topics such as menstruation, contraception, pregnancy, childbirth, and menopause, and prepare better questions before meeting a doctor. Antoniak et al. [30] presented an ethics- and stakeholder-centered framework for the use of natural language processing (NLP) and LLMs in maternal health care; rather than proposing a new predictive model, their study examines how stakeholders believe NLP tools should be designed and deployed. Adhikary et al. [31] explored menstrual health education by fine-tuning Meta-Llama-3-8B-Instruct on MENST, a new dataset of 23,820 menstrual health question–answer pairs, and reported that the resulting model provides accurate, easy-to-understand, and culturally sensitive responses. Shahnawaz et al. [32] proposed a human-centered, stigma-aware menstrual health chatbot designed for young women in Pakistan.

Table 1
Association between each feature and menstrual cycle phase (Kruskal–Wallis test for numerical features, chi-square test for categorical features). Features are grouped by type and sorted by p-value within each group; "Significant" means $p < 0.05$. †Below floating-point precision (reported as 0 by the test).

Feature	Type	p-value	Significant
PdG	Numerical	$< 10^{-300}$ †	Yes
Estrogen	Numerical	1.53×10^{-135}	Yes
LH	Numerical	7.07×10^{-113}	Yes
Flow volume	Categorical	$< 10^{-300}$ †	Yes
Flow color	Categorical	$< 10^{-300}$ †	Yes
Cramps	Categorical	9.92×10^{-81}	Yes
Sore breasts	Categorical	1.11×10^{-41}	Yes
Bloating	Categorical	3.46×10^{-19}	Yes
Indigestion	Categorical	2.23×10^{-7}	Yes
Headaches	Categorical	2.32×10^{-7}	Yes
Mood swings	Categorical	3.73×10^{-6}	Yes
Food cravings	Categorical	4.97×10^{-4}	Yes
Sleep issues	Categorical	9.60×10^{-3}	Yes
Fatigue	Categorical	1.24×10^{-2}	Yes
Appetite	Categorical	6.83×10^{-2}	No
Exercise level	Categorical	7.28×10^{-2}	No
Stress	Categorical	2.26×10^{-1}	No

2.3. On-Device Small Language Models

Small language models (SLMs) have emerged as a practical alternative to cloud-hosted large language models for consumer devices, as local execution reduces latency and cloud costs while keeping sensitive data on the device [16, 33, 34]. In this context, Lu et al. [35] surveyed open-source decoder-only SLMs with 100M–5B parameters and evaluated their on-device inference latency and memory requirements. Similarly, Liu et al. [33] demonstrated that sub-billion-parameter models using deep-and-thin architectures, embedding sharing, and grouped-query attention can support common on-device applications. More recently, compact general-purpose models, including Phi-3-mini [36] and the edge-oriented 0.5B, 1.5B, and 3B variants of Qwen2.5 [37], have further narrowed the performance gap with larger models. For instance, the 3.8B-parameter Phi-3-mini requires approximately 1.8 GB of memory with 4-bit quantization and generates more than 12 tokens per second entirely offline on an iPhone 14 [36]. These advances are supported by model compression techniques such as activation-aware weight quantization (AWQ), which reduces model weights to 4 bits and accelerates inference on desktop and mobile GPUs [34]. In addition, QLoRA enables efficient fine-tuning by propagating gradients through a frozen 4-bit model to train low-rank adapters [38]. Despite these improvements, studies on smartphones and edge devices indicate that memory constraints, energy consumption, and thermal limitations continue to affect sustained on-device execution [17, 39].

While these studies demonstrate the feasibility of deploying SLMs on resource-constrained devices, adapting these models locally remains challenging. To address this limitation, FwdLLM [40] combines backpropagation-free training with parameter-efficient fine-tuning to support federated fine-tuning of billion-parameter language models on commodity mobile devices. Meanwhile, in the healthcare domain, Health-LLM [41] demonstrated that fine-tuned language models can predict health outcomes from wearable sensor data across ten tasks involving mental health, physical activity, metabolic health, and sleep assessment.

2.4. Federated Learning for Privacy in Women's Health Applications

Federated learning offers a promising way to improve privacy in women's health applications by enabling model training without storing sensitive personal data centrally. In women's health settings such as menstrual tracking, symptom monitoring, fertility support, and personalized feedback generation, user data often include highly private physiological, behavioral, and self-reported information. Pavlidis et al. [42] proposed a privacy-friendly ovulation-tracking app that combines federated learning, privacy-enhancing technologies, and blockchain to protect sensitive reproductive health data while still providing predictive support. Rajesh [43] proposed an AI-powered menstrual cycle tracking system that combines contactless biosensing with federated learning (FL) to improve ovulation and menstrual phase prediction while preserving user privacy.

Summary: Although prior work has explored privacy-preserving ovulation tracking with federated learning, to the best of our knowledge no existing approach integrates federated learning with SLMs to enable privacy-aware, personalized feedback generation from menstrual health data.

3. Proposed Framework

We first describe the mcPHASES dataset and our statistical analysis, including missing-value handling and feature–phase association testing. We then present the proposed federated SLM-based framework for privacy-preserving menstrual phase prediction and personalized feedback generation.

3.1. Dataset and Statistical Analysis

Dataset: The mcPHASES dataset [44] (menstrual cycle Physiological, Hormonal, and Self-Reported Events and Symptoms) is a multimodal dataset designed to support research on menstrual health tracking with wearable and self-reported data. It contains de-identified data from 42 Canadian young adult menstruators collected over two three-month intervals. The dataset comprises 5,659 records, each with 17 features and a menstrual phase label (menstrual, follicular, fertility, or luteal). The features include three numerical hormone measurements (LH, estrogen, and PdG) and 14 categorical features describing menstrual flow, physical symptoms, and lifestyle factors.

Statistical Analysis: All 5,659 records contain at least one missing value, with the highest missingness in PdG, flow volume, and flow color. For the statistical analysis, we impute missing LH, estrogen, and PdG values with phase-wise medians and replace missing categorical values with `not_reported`. Because phase-wise medians depend on the phase label, this imputation is used only for the statistical tests and not for model training (see Step 2). We then apply the Kruskal–Wallis test to the numerical features and the chi-square test to the categorical features to examine feature–phase associations. As shown in Table 1, all features except appetite, exercise level, and stress are significantly associated with menstrual phase at $\alpha = 0.05$ (without correction for multiple comparisons).

3.2. Federated SLM-Based Menstrual Health Intelligence

Framework Overview

Goals: This section describes the federated learning framework used for menstrual health prediction, illustrated in Fig. 2 and Algorithm 1. The framework has two objectives: first, to protect sensitive user data while providing meaningful, personalized feedback; and second, to remain lightweight enough for deployment on resource-constrained consumer devices.

Workflow: The framework consists of the following steps. First, clients are selected and their data are partitioned at the subject level. Each client then preprocesses its data

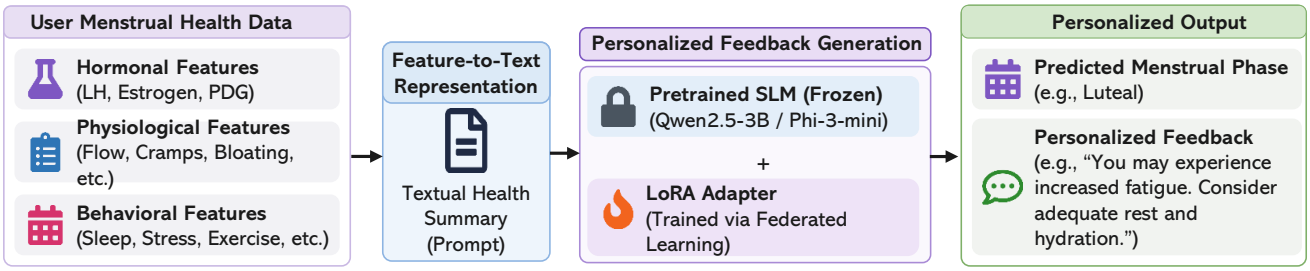


Figure 3: Personalized menstrual phase prediction and feedback generation using a federated LoRA-adapted SLM.

locally and converts each record into text for the SLM. Next, prompts are constructed for joint menstrual phase prediction and feedback generation. The prompts are tokenized for supervised fine-tuning, and LoRA is used for efficient model adaptation. Local client training is then performed within the federated learning setting, with periodic aggregation of model updates, validation, and checkpointing. Finally, during inference, the trained model generates a phase prediction and the corresponding personalized feedback for each user.

Evaluation metrics: We use two categories of metrics. The first measures predictive performance using accuracy, macro F1-score, and weighted F1-score. These metrics are defined as follows:

$$\text{Accuracy} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{I}(y_i = \hat{y}_i), \tag{1}$$

where N is the number of evaluated samples, y_i and $\hat{y}_i$ are the true and predicted phase labels, respectively, and $\mathbb{I}(\cdot)$ is the indicator function. For each phase c, the F1-score is computed as

$$\text{F1}_c = \frac{2\text{TP}_c}{2\text{TP}_c + \text{FP}_c + \text{FN}_c}, \tag{2}$$

where TP_c, FP_c, and FN_c denote the true positives, false positives, and false negatives for phase c, respectively. Scores with a zero denominator are assigned a value of zero. The macro and weighted F1-scores are then calculated as

$$\text{Macro-F1} = \frac{1}{C}\sum_{c=1}^{C} \text{F1}_c, \qquad \text{Weighted-F1} = \sum_{c=1}^{C} \frac{n_c}{N}\text{F1}_c, \tag{3}$$

where $C = 4$ is the number of phases and n_c is the number of evaluated samples belonging to phase c. Macro F1 weights all four phases equally, making it sensitive to performance on less frequent phases, whereas weighted F1 weights each phase according to its frequency. The generated feedback is assessed qualitatively through representative examples (Table 3).

The second category measures system performance during inference: CPU usage, process RAM usage, RAM increase during inference, system RAM usage. These metrics indicate the feasibility of deployment on resource-constrained devices.

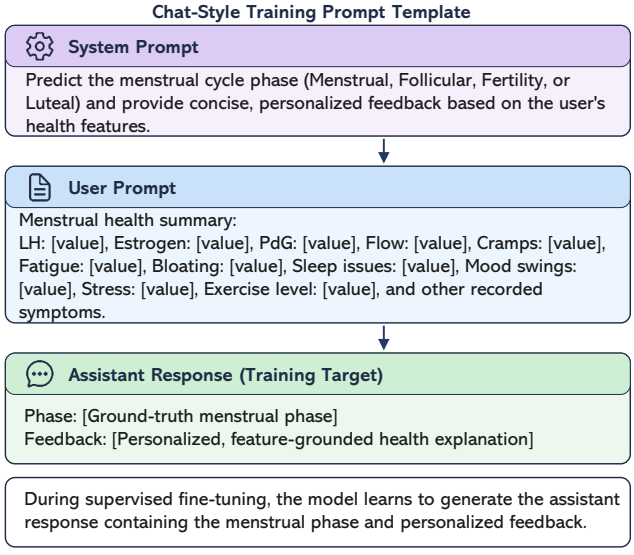


Figure 4: Chat-style training prompt template for menstrual phase prediction and personalized feedback generation.

Framework Steps

Step 1 *(Client Selection and Subject-Level Data Partitioning):* Let the client set be denoted by $C = \{c_1, c_2, \ldots, c_M\}$, where each client c_i corresponds to one participant. We use data from a subset of $M = 15$ participants as the client set C. The data are partitioned at the subject level, so that all records of a participant belong to only one subset: 10 clients are used for federated training, 2 for validation, and 3 for testing.

Step 2 *(Feature-to-Text Representation):* Before conversion, the phase label and the client identifier are removed from the model input. Missing numerical values are filled with median imputation and missing categorical values with the most frequent category, without using the phase label. Each record is then converted into a short textual health summary in which the numerical and categorical features are written as natural-language key–value descriptions, preserving the meaning of the original hormone values, symptom indicators, and self-reported information. This allows the model to read structured health measurements as text.

Step 3 *(Prompt Design for Phase Prediction and Feedback Generation):* We use a chat-style format to fine-tune the

Algorithm 1 Federated SLM Fine-Tuning for Menstrual Health Intelligence

Input: client set C, local datasets $\{D_i\}$, initial adapter θ^0, communication rounds T, local epochs E
Output: trained global adapter θ^T
Initialize LoRA adapter θ^0 and keep the pretrained SLM parameters fixed
for each client $c_i \in C$ **do**
    Convert menstrual health records into prompt–response pairs (phase prediction + feedback)
    Tokenize inputs and mask prompt portions to focus training on responses
end for
for each round $t = 0$ to $T - 1$ **do**
    Select a subset of clients S_t for participation
    for each client $c_i \in S_t$ **do**
        Load current global adapter θ^t
        Perform local fine-tuning on D_i for E epochs using LoRA
        Send updated adapter θ_i^{t+1} to the server
    end for
    Aggregate client updates using weighted averaging based on data size:
        $\theta^{t+1} = \sum_{i \in S_t} \frac{n_i}{\sum_{k \in S_t} n_k} \theta_i^{t+1}$
end for
for each test sample x **do**
    Generate predicted phase $\hat{y}$ and corresponding personalized feedback $\hat{f}$
end for
return final adapter θ^T

SLMs for both classification and explanation generation. As shown in Fig. 4, each sample consists of a system prompt, a user prompt, and an assistant response. The user prompt contains the feature values listed in Table 1, and the system prompt instructs the SLM to predict the menstrual cycle phase and generate feedback.

The assistant response, which serves as the training target, contains the ground-truth phase label and personalized feedback. The target feedback is constructed from the phase label and the features reported in each record, so that it refers only to the features present in that record. For example, bleeding-related indicators are emphasized in menstrual records, elevated LH and estrogen values contribute to fertility-related explanations, and symptoms such as fatigue, headaches, bloating, sore breasts, sleep issues, and mood changes are mentioned whenever they are reported.

Step 4 *(Supervised Fine-Tuning Data Construction):* After constructing the prompts, each example is tokenized with the model's tokenizer. Prompt tokens are masked out of the loss so that training focuses on the assistant response, which encourages the model to learn to generate the phase prediction and feedback rather than to reproduce the prompt.

Step 5 *(Base Model and Parameter-Efficient Adaptation):* We use Qwen2.5-3B-Instruct and Phi-3-mini-Instruct as base SLMs and train each separately. To reduce computational cost and memory requirements, each model is adapted with LoRA: the pretrained weights are frozen and low-rank adapters are added to the main transformer projection layers. Federated training runs for $T = 50$ communication rounds, and local optimization uses AdamW with a linear warm-up scheduler and gradient accumulation to improve training stability and memory efficiency. Keeping the number of trainable parameters small is especially useful in federated and resource-constrained settings.

Step 6 *(Local Client Training in Federated Learning):* As outlined in Algorithm 1, in each federated round t a subset of clients $S_t \subseteq C$ is selected. The server sends the current global LoRA adapter θ^t to each selected client $c_i \in S_t$, which fine-tunes it on its own dataset D_i for E local epochs. Let $n_i = |D_i|$ denote the number of local samples of client c_i. The local training objective is

$$\mathcal{L}_{c_i}(\theta) = \frac{1}{n_i} \sum_{(x_j, y_j) \in D_i} \ell\big(f_\theta(x_j), y_j\big), \tag{4}$$

where $f_\theta(\cdot)$ denotes the SLM with LoRA adapter parameters θ and $\ell(\cdot)$ is the training loss. After E local epochs, client c_i returns its updated adapter θ_i^{t+1}.

Step 7 *(Federated Aggregation):* After all selected clients finish local training, the server aggregates the returned adapters with federated averaging (FedAvg):

$$\theta^{t+1} = \sum_{i \in S_t} \frac{n_i}{\sum_{k \in S_t} n_k} \theta_i^{t+1}, \tag{5}$$

so that clients with more local data contribute more to the global adapter. The resulting θ^{t+1} is sent to the clients in the next round.

Step 8 *(Validation and Checkpointing):* After each round, the global adapter is evaluated on the validation clients, and it is saved together with the tokenizer whenever the validation loss improves.

Step 9 *(Inference and Output Generation):* During inference, the saved global adapter is attached to its base model (Qwen2.5-3B-Instruct or Phi-3-mini-Instruct). Each test record is converted into the same chat-style prompt used during training, and the model generates the predicted phase and personalized feedback. The generated text is parsed to extract the phase label and the feedback, so the system produces two outputs for every test record: a phase prediction and a natural-language explanation grounded in the input features.

Summary of the Framework: The framework combines subject-level client partitioning, feature-to-text conversion, federated LoRA fine-tuning, and personalized feedback generation (Fig. 3). Because only lightweight LoRA adapters are trained locally and shared, raw health records stay on the clients while training remains efficient.

Table 2
Prediction performance and inference resource usage of the federated SLMs (mean ± standard deviation for resource metrics).

Metric	Qwen2.5-3B	Phi-3-mini
Accuracy	**0.8394**	0.8251
Macro F1	**0.8179**	0.8032
Weighted F1	**0.8349**	0.8210
CPU Usage (%)	97.9 ± 1.3	95.1 ± 1.8
RAM (GB)	1.56 ± 0.05	1.46 ± 0.00
RAM Increase (MB)	5.3 ± 64.3	6.7 ± 66.6
System RAM (%)	65.8 ± 0.5	71.5 ± 6.6

4. Results and Evaluation

The federated experiments use the subject-level split described in Step 1, with 10 training, 2 validation, and 3 test clients. All experiments were conducted on a workstation with 90 GB of RAM.

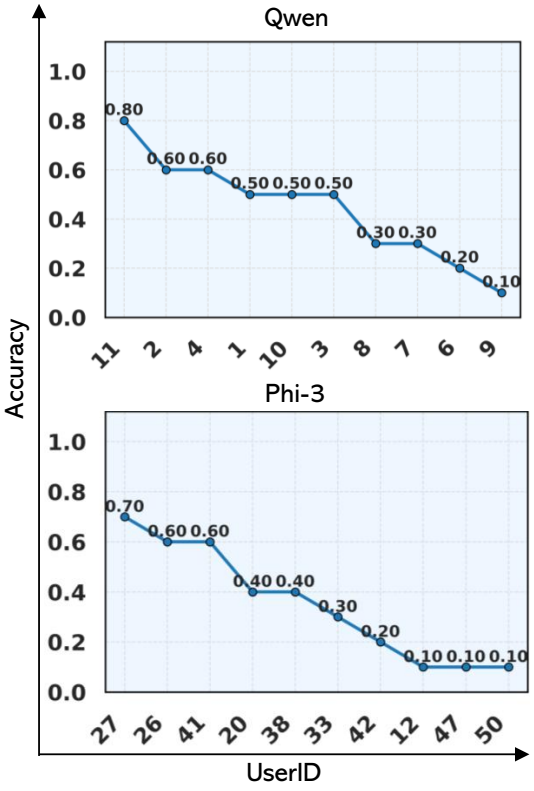


Figure 5: Per-user prediction accuracy of Qwen2.5-3B (top, test users 1–11) and Phi-3-mini (bottom, test users 12–50).

4.1. Results of the Federated SLMs

This section reports the results of Qwen2.5-3B-Instruct and Phi-3-mini-Instruct in the federated setting.

4.1.1. Results on Qwen2.5-3B-Instruct

As shown in Table 2, Qwen2.5-3B-Instruct achieves an accuracy of 0.8394, a macro F1-score of 0.8179, and a weighted F1-score of 0.8349. The macro F1-score is about 0.017 lower than the weighted F1-score, suggesting somewhat weaker performance on the less frequent phases.

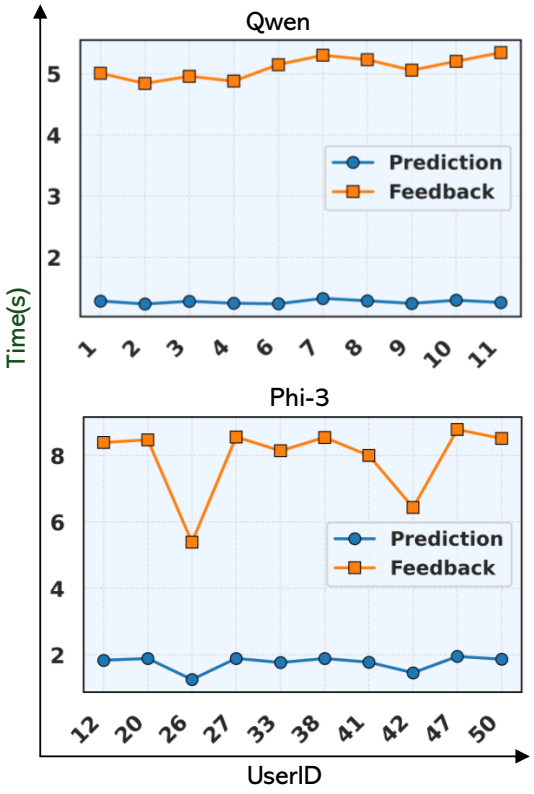


Figure 6: Per-user phase prediction time and feedback generation time (in seconds) of Qwen2.5-3B (top) and Phi-3-mini (bottom).

Figures 5–8 show user-level results. Accuracy ranges from about 0.10 to 0.80 across users (Fig. 5), indicating substantial user-specific differences in prediction performance. Prediction time stays around 1–1.3 s, whereas feedback generation takes about 4.8–5.4 s (Fig. 6). Input length (about 400–440 tokens; Fig. 7) and output length (about 31–34 tokens; Fig. 8) vary little across users, showing that the model processes inputs of similar length and produces feedback of stable size.

Table 3 shows representative, correctly classified test records with the feedback generated by Qwen2.5-3B. In most cases, the feedback is consistent with the predicted phase and refers to relevant patterns such as hormonal changes, cramps, or energy levels. Some responses are imprecise, however: for User 1 (menstrual phase), the feedback describes "a transition after menstruation", and for User 3 it states that the body "is preparing for menstruation". The responses also follow a small number of recurring templates. These observations motivate the systematic feedback evaluation discussed in Section 6.

In terms of resources (Table 2), Qwen2.5-3B uses 97.9% CPU and 1.56 GB of process RAM during inference. The average RAM increase during inference is small (5.3 MB), although its large standard deviation (64.3 MB) indicates fluctuations between runs. Deployment on smartphones or

wearables would require further compression, such as quantization.

4.1.2. Results on Phi-3-mini-Instruct

As shown in Table 2, Phi-3-mini-Instruct achieves an accuracy of 0.8251, a macro F1-score of 0.8032, and a weighted F1-score of 0.8210. These results are slightly lower than those of Qwen2.5-3B, with a similar gap between macro and weighted F1.

At the user level, accuracy varies noticeably, from about 0.10 to 0.70 (Fig. 5), indicating that prediction performance depends on user-specific data patterns. Prediction time remains low and stable at about 1.5–2 s, while feedback generation takes about 5.5–8.5 s, reflecting the higher cost of generating text (Fig. 6). Input length varies moderately between users (about 450–475 tokens; Fig. 7), and output length is nearly constant at about 41 tokens (Fig. 8), indicating uniform feedback length. Because Phi-3-mini was evaluated on a different set of users (IDs 12–50) from Qwen2.5-3B (IDs 1–11), the user-level results of the two models should not be compared directly.

The Phi-3-mini examples in Table 3 show that the generated feedback is phase-specific and refers to relevant symptoms such as cramps, fatigue, headache, breast tenderness, bloating, and sleep issues. In terms of resources, Phi-3-mini uses 95.1% CPU and 1.46 GB of process RAM, slightly less than Qwen2.5-3B, although its system RAM usage is higher (71.5% vs. 65.8%).

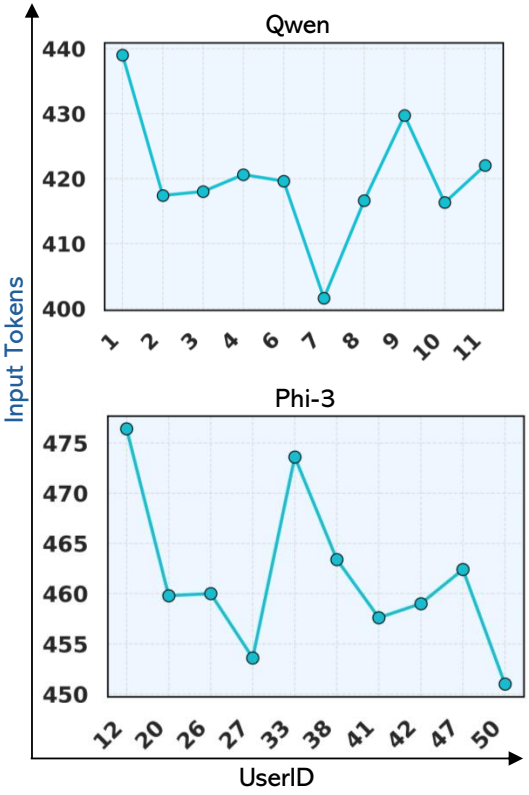


Figure 7: Per-user number of input tokens for Qwen2.5-3B (top) and Phi-3-mini (bottom).

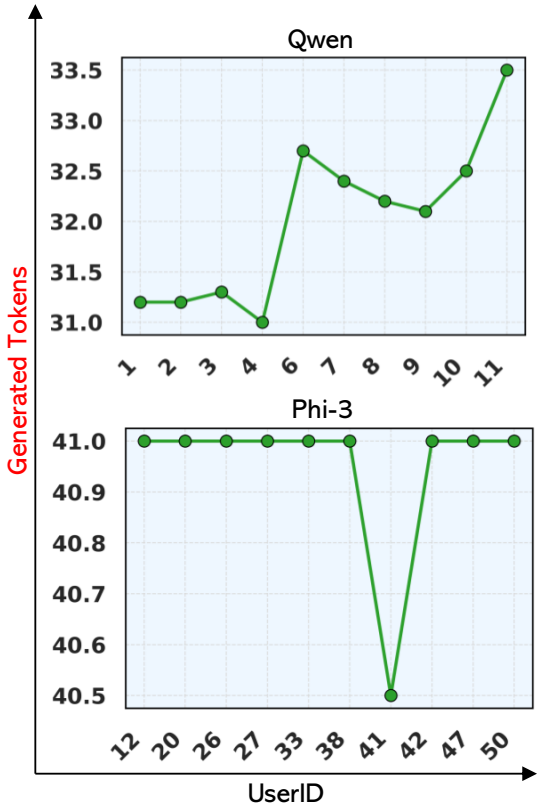


Figure 8: Per-user number of generated (output) tokens for Qwen2.5-3B (top) and Phi-3-mini (bottom).

Summary of Results: Both federated SLMs achieve accuracies above 0.82 and macro F1-scores above 0.80 while generating phase-specific feedback for users, with Qwen2.5-3B performing slightly better than Phi-3-mini. Both models also show large differences in accuracy between users, which points to the need for personalization strategies that better capture inter-individual variation. Phase prediction takes about 1–2 s and feedback generation about 5–8.5 s on a workstation, which is acceptable for non-interactive use such as daily summaries; real-time use on smartphones would require further optimization.

5. Pathway to Smartphone Deployment

Our implementation evaluates the proposed framework on workstation hardware; smartphone deployment is outside the scope of this study. Nevertheless, prior work provides evidence that both model families support on-device inference. Hao et al. [45] demonstrated quantized Qwen2.5-3B inference on commercial Android smartphones, reporting approximately 2.4 GiB of total memory consumption on a OnePlus 12 with a 4,096-token context budget. The Phi-3 technical report [36] demonstrated fully offline inference of 4-bit Phi-3-mini on an iPhone with an A16 Bionic chip at over 12 tokens per second. Furthermore, Nissen et al. [46] evaluated 4-bit Phi-3-mini-Instruct on multiple iPhones, including the iPhone 12 Pro, using an MLX-based inference application.

Table 3
Representative correctly classified test records and the feedback generated by Qwen2.5-3B and Phi-3-mini in the federated setting.

Model	User	True	Pred.	Generated Feedback
Qwen	1	Menstrual	Menstrual	The pattern suggests a transition after menstruation. Cramps seem notable; consider gentle self-care.
Qwen	2	Follicular	Follicular	The pattern suggests you are likely in the follicular phase. Your body is preparing for potential ovulation.
Qwen	3	Menstrual	Menstrual	The pattern suggests a transition into the menstrual phase. Your body is preparing for menstruation.
Qwen	4	Luteal	Luteal	The pattern suggests you are in the luteal phase. Your body seems to be preparing for a potential menstrual period.
Qwen	6	Menstrual	Menstrual	The pattern suggests a transition into the menstrual phase. Bleeding-related changes may be starting.
Qwen	7	Follicular	Follicular	The pattern suggests you are in the follicular phase. Your symptoms align with this time of your cycle.
Qwen	8	Luteal	Luteal	The luteal phase can be associated with changes in energy levels. Your current pattern may reflect typical luteal phase-related changes.
Qwen	9	Luteal	Luteal	The pattern suggests you are in the luteal phase. Hormone-related changes may be contributing to your pattern.
Qwen	10	Follicular	Follicular	The pattern suggests you are likely in your follicular phase. Your body seems to be preparing for potential ovulation.
Qwen	11	Follicular	Follicular	The pattern suggests you are likely in the follicular phase. Your symptoms seem to align with this time of your cycle.
Phi-3	12	Follicular	Follicular	The reported pattern is most consistent with the follicular phase. Hormone-related changes may be contributing to the pattern.
Phi-3	20	Menstrual	Menstrual	The reported pattern is consistent with the menstrual phase. Cramps may be contributing to discomfort.
Phi-3	26	Luteal	Luteal	The reported pattern is most consistent with the luteal phase. Breast tenderness may be relevant.
Phi-3	27	Menstrual	Menstrual	It appears you may be experiencing typical menstrual-related discomfort. Headache relief strategies may help.
Phi-3	33	Follicular	Follicular	The reported pattern is most consistent with the follicular phase. Hormone-related changes may be contributing to the pattern.
Phi-3	38	Luteal	Luteal	The reported pattern is most consistent with the luteal phase. Fatigue may be contributing to discomfort.
Phi-3	41	Menstrual	Menstrual	The reported pattern is consistent with the menstrual phase. Cramps may be contributing to discomfort.
Phi-3	42	Luteal	Luteal	It appears you may be experiencing bloating common in the luteal phase. Fatigue may be contributing to sleep issues.
Phi-3	47	Follicular	Follicular	The reported pattern is most consistent with the follicular phase. Hormone-related changes may be contributing to the pattern.
Phi-3	50	Menstrual	Menstrual	The reported pattern is consistent with the menstrual phase. Fatigue may be contributing to sleep issues.

These demonstrations support a potential deployment pathway in which the federated adapter is incorporated into a quantized model for local phase prediction and feedback generation. Accordingly, this study focuses on federated adaptation, predictive performance, and feedback generation rather than mobile runtime integration. The accuracy, latency, memory consumption, and thermal behavior of our adapted models after quantization remain to be validated on smartphones. The cited demonstrations establish inference feasibility, but do not establish the feasibility of local federated fine-tuning on these devices.

6. Limitations and Future Work

This study has several limitations. First, the federated setting is evaluated on a small number of clients, which may limit generalizability to diverse populations. Second, we rely on a single, recently released dataset that is limited to menstrual health tracking and contains substantial missing values; the required imputation and categorical replacement may affect data fidelity, and the approach has not yet been validated on other women's health applications. Third, the framework relies on feature-to-text serialization, which supports language-based feedback but may increase the risk of exposing sensitive information if prompts, logs, or model artifacts are not carefully protected. Fourth, although federated learning avoids centralizing raw data, the current framework does not include stronger privacy mechanisms such as secure aggregation or differential privacy. Fifth, the evaluation focuses on predictive performance and on inference resources measured on a workstation GPU; it does not quantitatively assess the faithfulness, readability, safety, or clinical usefulness of the generated feedback, nor resource usage on actual mobile or wearable hardware.

Future work will evaluate the framework on larger and more diverse women's health datasets, integrate secure aggregation and differential privacy, and assess the generated feedback through expert review and user studies. We also plan to explore multimodal extensions that integrate wearable, physiological, and self-reported data more directly, personalized federated adaptation strategies that better capture inter-individual variation, and model compression for deployment in a real-world mobile application.

7. Conclusion

In this paper, we presented a privacy-preserving and interpretable women's health support framework based on federated small language models for menstrual phase prediction and personalized feedback generation. The framework converts multimodal menstrual health data, including hormonal, physiological, and symptom-related features, into text-based prompts that two SLMs, Qwen2.5-3B-Instruct and Phi-3-mini-Instruct, use to predict the menstrual cycle phase and generate natural-language feedback. Trained with federated LoRA fine-tuning, so that raw health records never leave the clients, the models achieve accuracies of 0.839 and 0.825, respectively, and generate phase-specific feedback. We also evaluated the system's inference latency, CPU usage, memory use, and GPU memory consumption, providing

a basis for understanding the trade-offs between efficiency and model performance in consumer device deployments. Together, these results lay a foundation for combining personalized feedback with privacy-preserving learning in consumer health applications.

Declaration of generative AI

During the preparation of this work, the authors used ChatGPT (OpenAI) in order to check grammar and improve the readability of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Code availability

The source code will be made publicly available upon acceptance.

Data availability

This study uses the mcPHASES dataset [44], which is available on PhysioNet under restricted access (https://doi.org/10.13026/zx6a-2c81). Access requires registration and agreement to the dataset's data use terms; the authors do not have permission to redistribute the data.

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